Catalog of Questions MdamV

(Multidimensional Assessment for Visualization Understanding)

Please note that MdamV was originally developed and tested in German. For this publication, all questions were translated into English, but this language adaptation has not been tested. The question catalog reflects the version as of March 2023.

START OF SURVEY

Dear participant,

with this questionnaire, you are taking part in a research project of the Digital Humanities Master's program at the University of Vienna. The project deals with the **understanding of data visualizations**. In this questionnaire, your personal experience while dealing with two visualizations is of interest.

No prior knowledge is required. The questionnaire part for each of the two visualizations can be completed in about 5 minutes. Please make sure you answer every question. Exactly one answer can be selected for each question, unless there is an option for open answers in text form.

This project was funded by the Vienna Science and Technology Fund (WWTF) and the City of Vienna through the ICT20-065 project. Your questionnaire will not be passed on to third parties.

Thank you for your participation!

Continue.....1 == Continue

CONSENT

QCa I hereby confirm that I have read and understood this text and that I am over 16 years of age.

Consent given.....1 == Start of survey

DEMOGRAPHICS

Please fill in the following details.

QDa Age in years

Below 18.....1 == End of survey
18 to 24.....2
25 to 34.....3
35 to 44.....4
45 to 54.....5
55 to 64.....6
65 to 74.....7
75 or above.....8

QDb Gender

Male.....1
Female.....2
Diverse.....3
Inter.....4
Open.....5
Without gender entry.....6
I do not wish to make a specification here.....7

QDc Is your place of residence in Austria?

Yes.....1
No.....2 == End of survey

QDd Highest education level

No compulsory education.....1
Compulsory school certificate.....2
Baccalaureate.....3
University of applied sciences.....4
University degree.....6
Apprenticeship or other specialized training.....7
Master craftsman degree.....8
My degree is not listed. ….....9

TRANSITION

Climate change is shaping our current society and raises many questions about the future.

Data visualizations are often used to provide information about climate change. In this questionnaire you will see two visualizations on fact-based aspects of climate change.

Please look at the two visualizations and pay attention to both their content and your overall impression of the figure.

Continue.....1 == Continue

UNDERSTANDING OF VISUALIZATION 1

Please look at the visualization and continue when you have grasped the message of the figure.
You will still be able to see the illustration on the following pages of the questionnaire.
You can enlarge or reduce the visualization by zooming in or out.
[show V1]

Continue.....1 == Continue

Please assess the extent to which the statements apply.
The visualization can be found at the bottom of the page.
[show V1 beneath items]

Q1a-V1* The visualization seems understandable to me. The information can be clearly read.
Q1b-V1 I can classify the information presented in my previous knowledge and make connections.
Q1c-V1 I could summarize the content of the visualization for someone.

I disagree.....1
I somewhat disagree.....2
Undecided.....3
I somewhat agree.....4
I agree.....5

Please assess the extent to which the statements apply.
[show V1 beneath items]

Q2a-V1 I can understand the units of measurement shown here, e.g., millimeters.
Q2b-V1 I can easily see whether the colors used in the visualization show any specific information.
Q2c-V1 I am familiar with the line chart and it is not the first time that I have read information from such a chart.

I disagree.....1
I somewhat disagree.....2
Undecided.....3
I somewhat agree.....4
I agree.....5

Please assess the extent to which the statements apply.
[show V1 beneath items]

Q3a-V1 This visualization appeals to me.
Q3b-V1 This visualization is pretty to look at.
Q3c-V1 This visualization is well designed.
Q3d-V1 This visualization is trustworthy.

I disagree.....1
I somewhat disagree.....2
Undecided.....3
I somewhat agree.....4
I agree.....5

Please assess the extent to which the statements apply.
[show V1 beneath items]

Q4a-V1 I have found an error in the visualization.
Q4b-V1 I have a suggestion for a change to the visualization.
Q4c-V1 If I could, I would design something in the visualization differently.

I disagree.....1
I somewhat disagree.....2
Undecided.....3
I somewhat agree.....4
I agree.....5

* **V1** stands for Visualization 1. The illustration tested as V1 is shown on the last page of this catalog.

Q4d-V1 I would like to remark something positive about the visualization:
Q4e-V1 I would like to remark something negative about the visualization:

Text field (max. 2000 characters, optional) .....TEXT

Please decide whether the statements are true or false according to what the visualization shows.
[show V1 beneath items]

Q5a-V1 The sea level fluctuates, showing no trend.
True.....0 == Incorrect
False.....1 == Correct

Q5b-V1 In 2021, the sea level has risen by around ten centimeters compared to 1993.
True.....1 == Correct
False.....0 == Incorrect

UNDERSTANDING OF VISUALIZATION 2

Please look at the visualization and continue when you have grasped the message of the figure.
You will still be able to see the illustration on the following pages of the questionnaire.
You can enlarge or reduce the visualization by zooming in or out.
[show V2]

Continue.....1 == Continue

Please assess the extent to which the statements apply.
The visualization can be found at the bottom of the page.
[show V2 beneath items]

Q1a-V2 The visualization seems understandable to me. The information can be clearly read.
Q1b-V2 I can classify the information presented in my previous knowledge and make connections.
Q1c-V2 I could summarize the content of the visualization for someone.

I disagree.....1
I somewhat disagree.....2
Undecided.....3
I somewhat agree.....4
I agree.....5

Please assess the extent to which the statements apply.
[show V2 beneath items]

Q2a-V2 I can understand the units of measurement shown here, e.g., millions of tons.
Q2b-V2 I can easily see whether the colors used in the visualization show any specific information.
Q2c-V2 I am familiar with the bar chart and it is not the first time that I have read information from such a chart.
I disagree.....1
I somewhat disagree.....2

Undecided.....3
I somewhat agree.....4
I agree.....5

Please assess the extent to which the statements apply.
[show V2 beneath items]

Q3a-V2 This visualization appeals to me.
Q3b-V2 This visualization is pretty to look at.
Q3c-V2 This visualization is well designed.
Q3d-V2 This visualization is trustworthy.

I disagree.....1
I somewhat disagree.....2
Undecided.....3
I somewhat agree.....4
I agree.....5

Please assess the extent to which the statements apply.
[show V2 beneath items]

Q4a-V2 I have found an error in the visualization.
Q4b-V2 I have a suggestion for a change to the visualization.
Q4c-V2 If I could, I would design something in the visualization differently.

I disagree.....1
I somewhat disagree.....2
Undecided.....3
I somewhat agree.....4
I agree.....5

Q4d-V2 I would like to remark something positive about the visualization:
Q4e-V2 I would like to remark something negative about the visualization:

Text field (max. 2000 characters, optional) .....TEXT

Please decide whether the statements are true or false according to what the visualization shows.
[show V2 beneath items]

Q5a-V2 Only Russia and Germany show a decrease in CO_2 emissions over the 28 years shown.
True.....1 == correct
False.....0 == incorrect

Q5b-V2 Japan had the lowest increase in CO_2 emissions in 2018 compared to 1990.
True.....0 == incorrect
False.....1 == correct

Please answer the following question.

Q5c-V2 Which country is listed first in the visualization?

Text field (max. 2000 char., obligatory.....TEXT, „China“ (or similar) == correct

Please assess the extent to which the statements apply.

QDe I deal a lot with the topic of climate change out of my own interest.
QDf I have a good knowledge of mathematics, especially in the field of statistics.

I disagree.....1
I somewhat disagree.....2
Undecided.....3
I somewhat agree.....4
I agree.....5

FEEDBACK

QFa Any other feedback or comments you would like to make on this questionnaire:

Text field (max. 2000 characters, optional) .....TEXT

CLOSING

Thank you for your participation!
Your answers have been saved. You can now close this browser window.

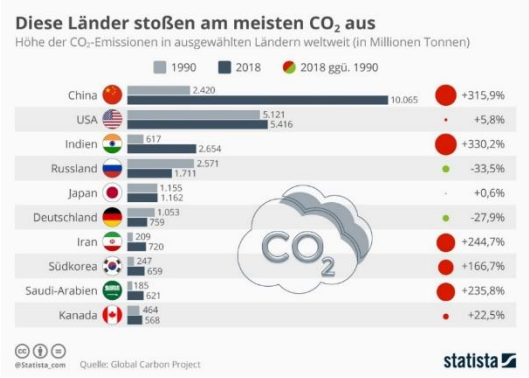
SURVEYED CLIMATE DATA VISUALIZATIONS

For a description of the data visualization's content and visual features in English please refer to Section 2.3 in the thesis.

V1: A line chart on sea level data.

(Source: Statista, *Der Meeresspiegel steigt kontinuierlich an*, NASA, https://de.statista.com/infografik/21922/anstieg-des-meeresspiegels/)

V2: A bar chart showing selected countries' CO_2 emissions over time.



(Source: Statista, *Diese Länder stoßen am meisten CO aus*, Global Carbon Project, URL: https://de.statista.com/infografik/18287/co2-emissionen-in-ausgewaehlten-laendern/)